

Zakaria Tikialine

Software Engineer | Full Stack Developer | Associate Data Scientist

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Professional Summary

Full Stack Software Engineering student with hands-on experience designing and deploying production-ready web systems and data-driven applications. Strong backend orientation with proven ability to build scalable REST APIs, optimize relational databases, and integrate machine learning models into practical solutions. Certified Associate Data Scientist with solid foundations in predictive modeling and statistical evaluation.

Technical Skills

Programming: Python, TypeScript, JavaScript, Golang, SQL

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Backend: Node.js, Express.js, REST APIs, WebSockets, Authentication, RBAC

Databases: PostgreSQL, MongoDB, Oracle

Machine Learning: Scikit-learn, Regression, Classification, Feature Engineering, Model Evaluation

Data Analysis: Pandas, NumPy, Matplotlib

DevOps & Tools: Git, Docker, Linux, Vercel, Railway

Languages: Arabic (Native), French (B2 – TCF), English (Professional)

Professional Experience

NAFTAL – Direction Centrale des Systèmes d'Information

Jun 2025 – Jul 2025 *Full Stack Web Developer Intern*

- Designed and deployed a full-stack employee travel management system supporting multiple organizational centers and seasonal campaigns.
- Architected secure RESTful APIs with role-based access control and structured authentication workflows.
- Modeled and optimized PostgreSQL relational schemas to ensure efficient querying and data consistency.
- Developed a responsive frontend using Next.js and TypeScript with clean state management and modular components.
- Automated participant selection logic and contributed to production deployment workflows.

School of AI Club – Bejaia

Jan 2026 – Present *Technical Assistant*

- Assisted in organizing and delivering workshops on Machine Learning, Deep Learning, and backend development.
- Mentored participants during hands-on sessions involving Python, data preprocessing, and model training.

Nexus Security Club – ESTIN

Aug 2024 – Aug 2025 *Logistics Manager*

- Coordinated logistics and technical resources for cybersecurity competitions involving 75+ participants.
- Collaborated with cross-functional teams to ensure smooth execution of technical events.

Personal Projects

Student Performance Prediction Model

(Python, Scikit-learn)

- Designed an end-to-end machine learning pipeline including data cleaning, feature engineering, model training, and validation.
- Conducted exploratory data analysis using Pandas and NumPy to identify key predictive factors.
- Implemented and compared regression models (Linear Regression, Lasso) achieving $R^2 > 0.70$.
- Applied cross-validation and error metrics (MSE) to ensure robust performance evaluation.

Tahwisa – Employee Registration Platform

(Next.js, Express.js, PostgreSQL)

- Built a secure authentication system with role-based access control.
- Designed normalized relational schemas and optimized database queries.
- Implemented automated selection algorithms and deployed the application in a production-ready environment.

Education & Certification

École Supérieure en Sciences et Technologies de l'Informatique (ESTIN) State Engineer in Computer Science – Artificial Intelligence & Data Science Sep 2022 – 2027

Associate Data Scientist – DataCamp Professional Certification